

Electricity Grids and Distributed Renewable Energy - Heat Pumps



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Introduction/Background

For the vast majority of human history, our ability to grow and advance has been directly tied to our access to energy. This has been observed in many different ways throughout the ages, from fire to light our dwellings and cook our food, watermills to process our resources, coal and later fossil fuels to spin our turbines and drive the wheels of our vehicles, and muchmore. In the present day, this phenomenon not only remains, but has become more prominent than ever.

Nuclear fission, photovoltaic cells, and wind turbines are just a few of the clever and futuristic ways that we have learned how to harness alternative energy generation methods for the sake of minimizing the harmful effects of more traditional fossil fuel based energy sources. However, with the advent of this new technology comes a bevy of new challenges that we must learn to navigate if we are to successfully integrate it into our greater society. Electrification of our appliances and transportation has been a focal point of policy makers and engineers (Sugiyama, 2012) alike since the dangers of anthropological climate change have been more properly realized, and while on paper it seems a simple enough concept to tackle the fact is that there is still much work to be done to fully understand the nuances of infrastructure, policy, and usage habits.

Many organizations both private and public are hard at work exploring the avenues available to us that may help facilitate this transition into a more electrified society, incorporating these new technologies in creative ways that help circumnavigate the challenges posed by finite resources and detrimental effects of using these finite resources. One such topic that is of great importance involves the methods and habits for the heating and cooling of our dwellings and other buildings.

Heating and cooling account for roughly half of the entire world's energy usage, with this energy being produced from a mixture of fossil fuels and electricity. While most of the cooling in the world is electrified, most heating is done via the usage of natural gas and other fossil fuels. Thus, the potential energy savings from climate control for heating will come from the electrification of the process, while for cooling it will be achieved from shifting usage habits and practices. Additional savings for heating will also be born from further optimization of usage patterns in a similar vein as cooling.

As previously stated, heating and cooling systems represent a significant portion of global energy demand in both residential and commercial sectors. In the residential sector, natural gas, wood, propane, and other fossil fuels are predominantly used for heating purposes due to their widespread availability and lower cost. Conversely, the commercial sector, characterized by larger scale buildings and variable occupancy, employs a more diverse range of energy sources including district heating, natural gas, and electric heat pumps. As of 2019, fossil fuels account for almost 55% of the energy used in the above heating methods around the world, with only 12% being powered by generated electricity. (IEA, 2019) Existing infrastructure and the substantial initial costs associated with retrofitting buildings or replacing systems with more energy-efficient alternatives pose significant barriers. Additionally, habitual behavior in energy usage such as the preference for constant indoor climate control irrespective of external temperatures compounds the difficulty of reducing energy consumption without sacrificing comfort or convenience.

Heating Pump/Air Conditioning Technology

Unlike fuel based heating, heat pumps transfer heat from colder to warmer spaces using a refrigeration cycle. They consist of a compressor and two coils (one inside and one outside), both containing refrigerant fluid. The compressor pumps the refrigerant between the coils, changing its pressure and state, thus allowing it to absorb and release heat to regulate temperatures in enclosed spaces. The adoption of heat pumps has been limited due to high upfront installation costs, the potential requirement for modifications to existing heating systems, and lower efficiency in regions with extremely cold climates. For cooling, traditional air conditioners use a similar refrigeration cycle but only in one direction, moving heat from the interior to the exterior. The limited adoption of more advanced cooling technologies often relates to the higher costs associated with these systems and the inertia of existing cooling solutions that are already effective in many climates.

This process is made possible through the wonders of thermodynamics, particularly those governing heat transfer and phase changes. The core principle that heat pumps and air conditioners leverage is the second law of thermodynamics, which states that heat naturally flows from a hotter to a cooler space. Using a refrigeration cycle, these devices artificially reverse this process. By compressing and expanding a refrigerant fluid, they force heat to move from a cooler area to a warmer one—opposite of its natural flow. This manipulation of heat flow allows for effective heating or cooling of spaces as required. The refrigerant fluid in heat pumps and air conditioners achieves heating and cooling through phase changes between liquid and gas states, driven by pressure changes. When compressed, the refrigerant heats up as it is pressurized in a gas state which then passes through a condenser where it loses heat and condenses into a

liquid. When the refrigerant is allowed to expand, it cools rapidly as it evaporates back into a gas in the evaporator coil, absorbing heat from the surrounding area. This cycle of compression, condensation, expansion, and evaporation enables the transfer of heat either into or out of a space, depending on the mode of operation of the device.

There also exists a wide variety of different heat pump types, some of which are more feasible than others depending on region. Ground source borehole heat pumps utilize deep underground pipes to exchange heat with the earth, leveraging the ground's constant temperature for high efficiency. These systems are ideal for areas where outdoor temperatures fluctuate widely. Ground water well heat pumps operate similarly but use groundwater extracted through wells as their heat exchange medium. This method requires sustainable water management to prevent aquifer depletion but offers efficient temperature control by utilizing the stable temperatures of groundwater. Air to water heat pumps extract heat from outdoor air to heat water, which is then used in the building's systems. This type is more sensitive to outdoor air temperatures and can be less efficient in extremely cold climates. Air to air heat pumps transfer heat directly between the indoor and outdoor air, functioning much like traditional air conditioners in reverse. These are less expensive to install and maintain but also vary in efficiency depending on the ambient air temperature.

Energy/Cost Savings

Due to the fact that heat pumps operate on electrical power, this would allow for more efficient power generation plants and methods to supply the electricity that will be used for domestic

heating purposes rather than directly utilizing fossil fuels in the residence/building. While the exact extent of energy and cost savings will vary from region to region, there exists clear potential in terms of reducing both emissions as well as energy costs for the consumer. A study of the United States and the potential savings it could see showed a clear relationship between the method of electricity generation and the associated savings. States with a large source of renewable generation and cooler climates saw as great as an 83% reduction in CO₂ emissions, while others that are still reliant on emission heavy generation like Montana saw an up to 65% increase in emissions. (Walker, 2021) Unfortunately the same study also concluded that energy savings are much more difficult to achieve due to the comparably low cost of fossil fuels, though the volatile cost of fossil fuels means that various scenarios exist where savings (though not as prominent as carbon savings) are a possibility. The age and efficiency of the heating systems they would replace was also a key factor in achieving a scenario where savings were predicted, with older and less efficient models having the greatest savings potential if replaced.

Another aspect that must be considered is the influence that different climates have on the efficiency of heat pump systems. Due to the way that heat pumps operate by pulling warm air from its colder surroundings, when conditions reach a certain point the ability for a heat pump to function drops considerably. This would mean that in locations that are not sufficiently warm, backup fossil fuel heating sources would be necessary for periods of extreme and intense cold. (Berardi, 2022) It is a stark reminder that while technological advances are able to bridge the gap between our current world and a carbon neutral one, there is no one size fits all solution.

While switching to a more energy efficient means of heating our homes is the first step to be taken in this process of reducing emissions and energy usage, there exist many more avenues for savings that can be done after the fact. The proliferation of smart meters installed in residential and commercial buildings is one such tool that has great potential in maximizing the savings that can be realized via intelligent usage and proper maintenance of the equipment to maximize energy and cost related savings. Researchers in Switzerland found that through the monitoring of heat pump based usage statistics, they could preselect locations to send technicians that would perform minor technical repairs and inform the operators of the equipment on how to best ensure that they understand the best practices related to smooth operation. An annual average savings of 1,805 kWh (or 15.2%) was observed when smart meters were used to preselect where to send the technicians, a substantial amount that when scaled up would lead to a significant reduction in wasted energy. (Weigert, 2022)

In addition to this type of monitoring that would allow for preemptive identification of households that would benefit from maintenance and upkeep, broader policy based initiatives also have the potential for significant energy savings. The demand curve is a well known energy usage phenomena that refers to the increased demand for electricity that can be observed at various points throughout the day. For example, in the afternoon during the hottest part of the day air conditioning will account for a sudden increase in demand that may be difficult for energy providers to meet. Another example would be that in communities that have high EV ownership rates, most owners would charge their vehicles overnight when they are home and not using them. This coupled with the fact that solar energy generation is intermittent means that at some times during the day energy must be curtailed, while at other times other sources of energy

must be utilized to make up for the difference.

Heating and cooling are both energy usage factors that are heavily influenced by these demand curves, as well as having many different ways in which savings could be realized. Research has already been conducted that analyzes the usage of heating and cooling for buildings when they are and are not occupied, showing that there are still many ways in which programmable “smart” thermostats would be able to minimize the usage of these appliances when there is no occupancy. One such experiment that attempted to reduce energy usage by creating a model that incorporated usage habits and weather patterns was able to achieve a 7% reduction in heat pump energy consumption, which may seem like a small number on paper but would add up to a significant amount when considering its effects on a larger population. (Mbuwir, 2024)

Obstacles

As with any other technology that has the potential to reduce our emissions, there remain many other fundamental issues that must be addressed beforehand. Much like with wide scale EV adoption, there is still insufficient electricity generation and distribution capacity to fully support large populations shifting toward full electrification. As a short example, the population of the United States is currently 330 million individuals. While there does not exist a database containing the number of heat pumps currently existing in the USA, we will make a liberal assumption that 20% of households have heat pumps installed. Total energy usage in the USA in 2022 amounted to 100.41 quadrillion BTU’s, or approximately 29.5 million GWh. Of this, 42% was used for space heating, which comes to approximately 12.3 million GWh. Based on our

previous assumption that 80% of heating energy is provided by fossil fuel based means, this amounts to 9.84 million GWh of energy.

Heat pumps are assigned a coefficient of performance (COP) value, which refers to the amount of heat energy that is released per energy consumed. The average heat pump has a COP of anywhere between 3.0 and 5.0, and using the conservative estimate of 3 would lead to a $\frac{2}{3}$ drop in overall energy used for heating. And although this would result in a final heating energy usage of ~2186664 GWh of energy, far less than the current amount, this amounts to roughly half of the total electricity generated by the USA for the year of 2023. This move to electrification would most likely be accompanied by a concurrent switch to EV vehicles and other small ways to switch out fossil fuels to cleaner sources, demand that we do not yet have the infrastructure capacity to meet. The nature of the demand is also quite variable, with one study finding a mere 6% increase in energy usage, but a 37% increase in peak demand in communities that switched to primarily heat pump based space heating. This means that the simple addition of generation capacity is not the greatest issue, but rather lies in figuring out how to ensure that these peaks and troughs are properly accounted for. (Knittel, 2024)

In addition to this, the cost of heating via electricity is still far greater than the costs we are able to achieve using fossil fuel based heating. Even at its most conservative estimate, heating costs more than double when done via electricity versus fossil fuel source, which would act as a significant psychological and financial barrier for households that may be considering the switch. Thankfully this switch can, and is being made cheaper through the installation of more renewable energy sources, more efficient electricity generation technology, and various

governmental subsidies. As previously touched on before, ambient climate plays a large role in the efficacy of heat pumps, which would suggest that policy would be most effective if shaped by geographical location and local climates.

Another factor to consider is that much like HVAC technology, heat pumps require specially trained technicians to properly diagnose and repair any mechanical or maintenance related problems that may arise. Currently there are not nearly enough technicians certified for this task in the United States or elsewhere, which may require the intervention of governmental entities to ensure that an adequate number of professionals are trained to meet future demand.

Policy and Incentivisation

As mentioned previously, while there exist a variety of obstacles that make widespread adoption difficult, there are a variety of ways in which local, state, and federal governments would be able to facilitate this transition. Subsidies and rebates similar to those seen for solar powered technologies are a low hanging fruit that would act as a financial incentive for adoption. Researchers have found that creating energy market models in conjunction with the implementation of this type of policy ensures that the grid's ability to meet this demand is not strained, and allows for more granular policy making that can maximize effectiveness. (Abd Alla, 2022) When combined with a more in depth understanding of what factors are of the highest likelihood to shape consumer behavior, these models have shown great promise in helping to take old, energy inefficient heating methods out of circulation and allowing for a more sustainable and energy conscious retrofitting process.(Strazzera, 2024)

Another way in which intelligent policy adoption can be supported is through concurrent integration with other related policies. More specifically, understanding how much the impact of this demand could be alleviated if battery storage systems, localized PV generation, and thermal storage are also taken into consideration when creating new policies. Studies have already been done that attempt to model different scenarios in which the above technologies work together to smooth the residential demand curve, which would allow for a greater margin of error throughout the electrification process. (Li, 2022) The aforementioned concern over the shortage of qualified technicians could also be alleviated via governmental education programs. Providing funding and incentivization to companies that are able to train tradesmen and women versed in heat pump operations would be a good start, funded through tax bills and green bonds.

Conclusion

The transition towards electrified heating systems represents a critical juncture in the pursuit of sustainable energy consumption. This shift is driven by the need to decrease reliance on fossil fuels and mitigate associated environmental impacts. Heat pumps, with their superior coefficient of performance relative to traditional heating methods, offer a promising solution for reducing both energy consumption and carbon emissions. However, the adoption of such technologies is not without challenges. The initial high costs of installation, coupled with the need for regular specialized diagnostic and maintenance pose substantial barriers. Furthermore, the effectiveness of heat pumps can diminish in extremely cold climates, necessitating supplemental heating sources that may offset some of the potential energy savings.

Additionally, the broader integration of heat pumps into the energy system involves complex interactions with existing electrical infrastructure. The potential increase in electricity demand, particularly at peak times, necessitates enhancements to grid flexibility and capacity. This is imperative not only to accommodate the increased load but also to ensure the reliability of energy supply in the face of fluctuating demand patterns. Policy interventions and incentives will play a pivotal role in facilitating this transition. Financial incentives such as subsidies and rebates can lower the entry barriers for consumers, while regulatory measures can enforce the gradual phase-out of less efficient heating systems. Moreover, strategic investment in technician training programs is essential to support the maintenance and optimization of heat pump installations. These policies will have a dual impact, as many of them have great potential in also benefiting our energy usage habits for cooling as well.

While the shift towards electrified heating via heat pumps is aligned with broader environmental goals and offers considerable benefits in terms of energy efficiency and emission reductions, it requires a coordinated approach involving technological, economic, and regulatory strategies. These must be tailored to local conditions to fully realize the potential of heat pump technology and ensure its successful integration into the energy landscape.

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